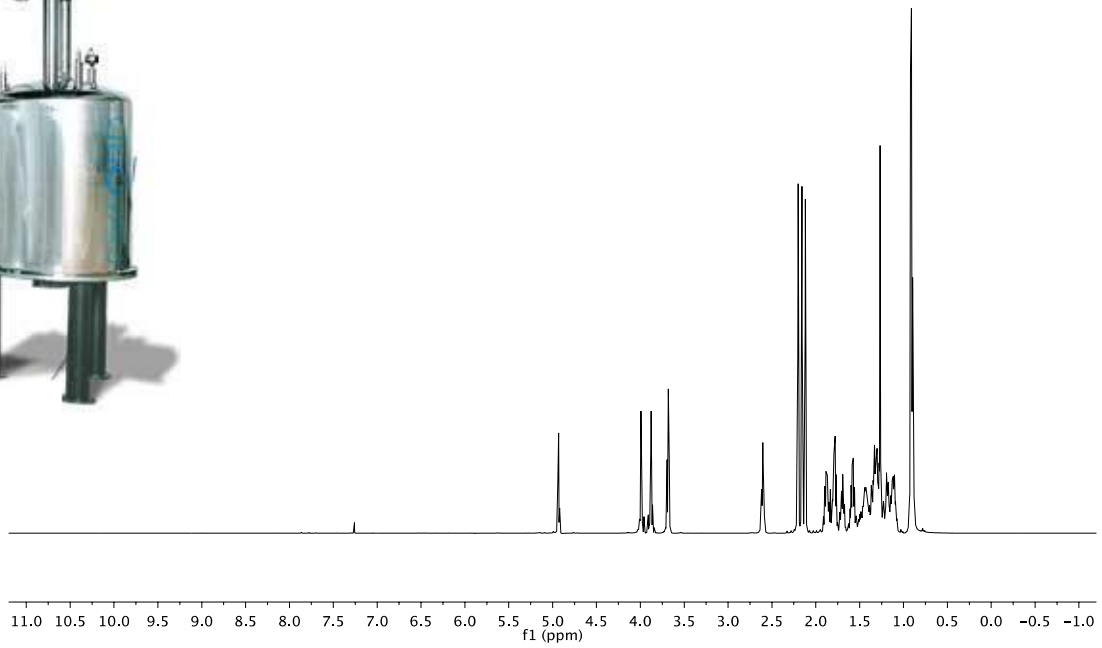
Nuclear Magnetic Resonance

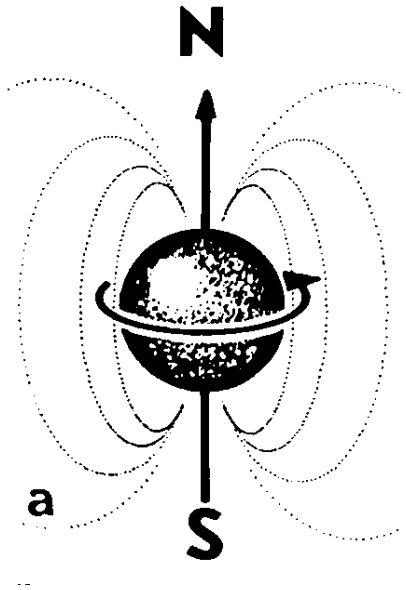


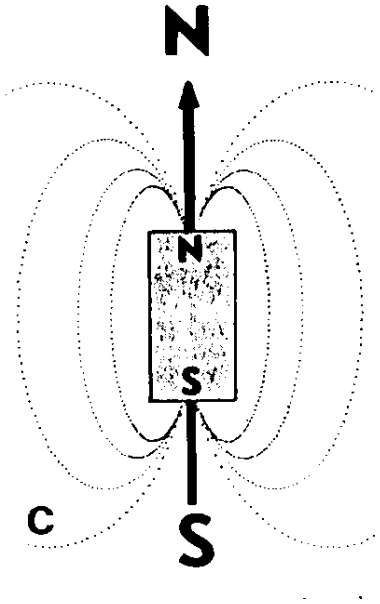


Dr J. L. Kiappes

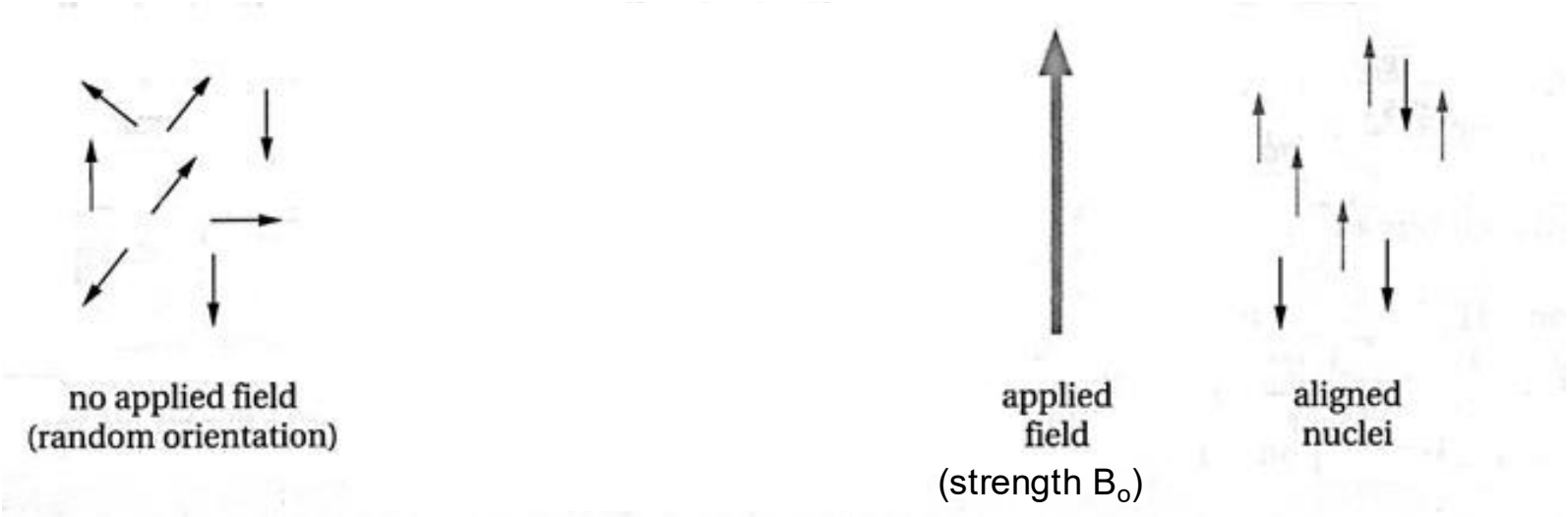
October 2026

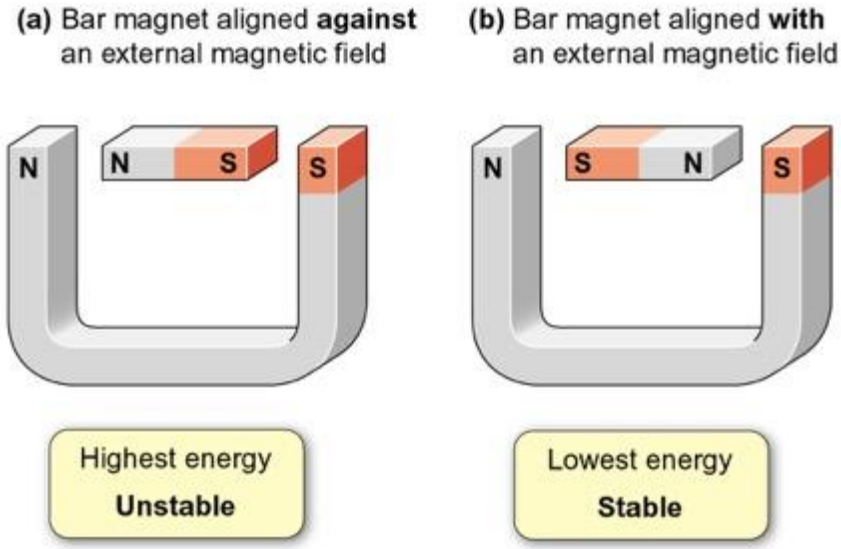
Nuclear Spins





Nuclear Spins in a Magnetic Field

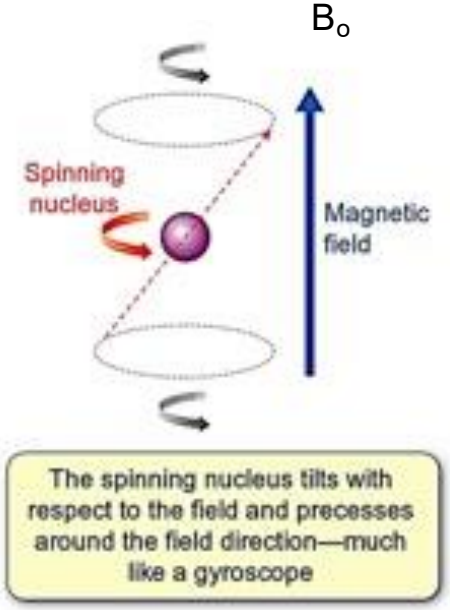




The nuclear spin axis in a magnetic field : The Larmor Frequency (ν_L)

$$\nu_L = \frac{\gamma B_o}{2\pi}$$





Different nuclei and their NMR (Larmor) frequencies in MHz for different applied magnetic field strengths

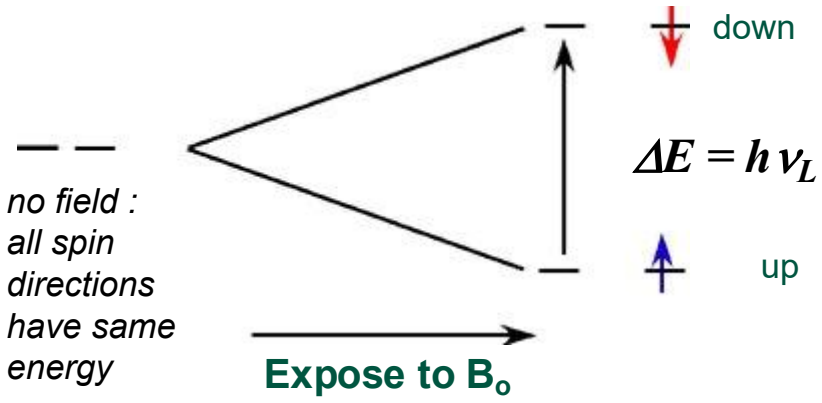
$$\nu_L = \frac{\gamma B_o}{2\pi}$$

1 MHz = 10^6 Hz (s^{-1})

Nucleus (isotope)	Larmor frequency if B_o = 11.7 T	Larmor frequency if B_o = 23.5 T	Spin (I)	Abundance (%)
^{1}H	500	1000 (1 GHz)	1/2	99.9885
^{13}C	126	252	1/2	1.07
^{15}N	50.7	101.4	1/2	0.364
^{31}P	202	404	1/2	100
^{19}F	470	940	1/2	100
^{17}O	67.8	135.6	5/2	0.038
^{23}Na	132	264	3/2	100
^{2}H (D)	76.8	153.5	1	0.0115

We will concentrate on NMR spectroscopy of ^{1}H nuclei, but the same principles apply to ^{13}C, ^{15}N, and other nuclei important in biological chemistry.

NMR Spectroscopy - A Quantum Mechanical Model

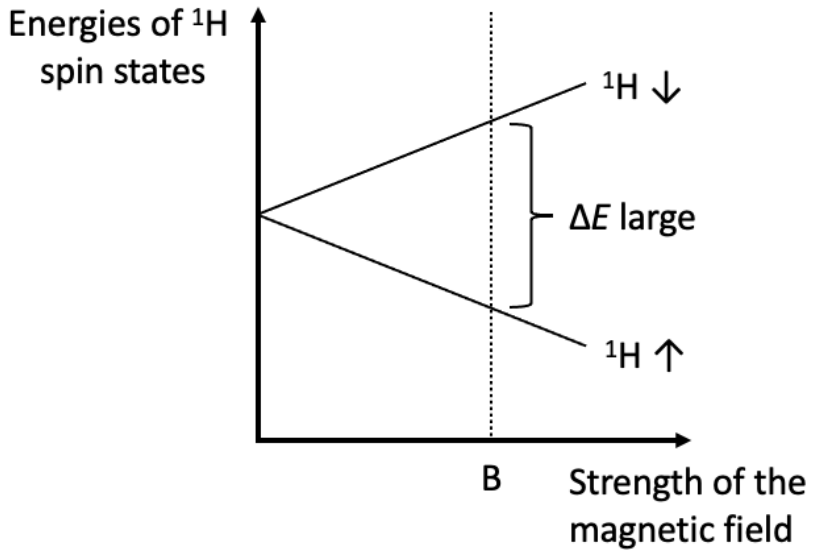


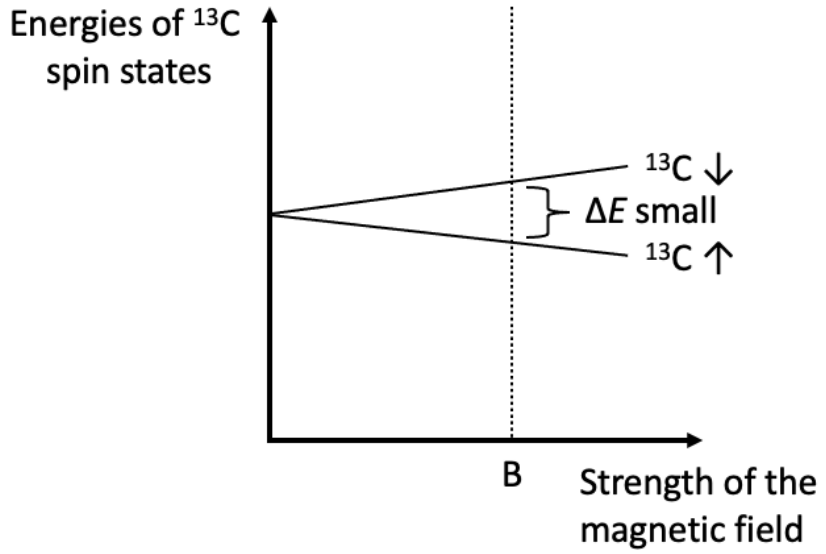
The energy difference between "spin up" (↑) and spin down" (↓) states depends on the applied magnetic field strength B_o

$$\Delta E = \frac{\gamma B_o h}{2\pi} = h\nu_L$$

Same Field, Different Nuclei

$$\Delta E = \frac{h\gamma B_0}{2\pi}$$



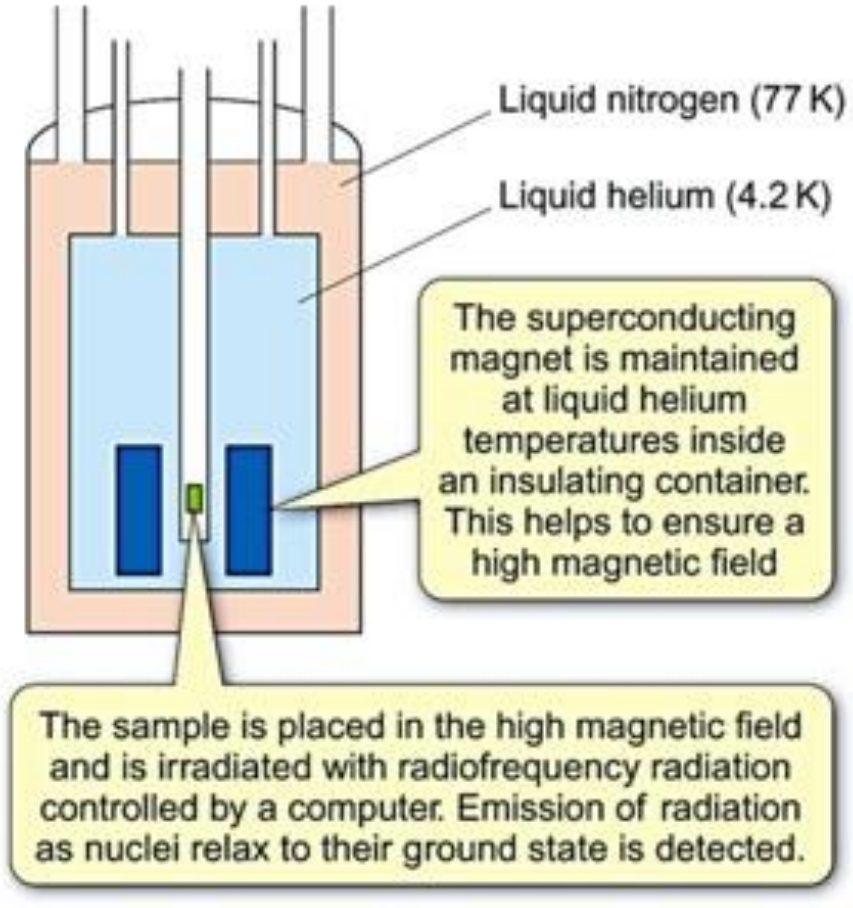


Carrying out an NMR Experiment



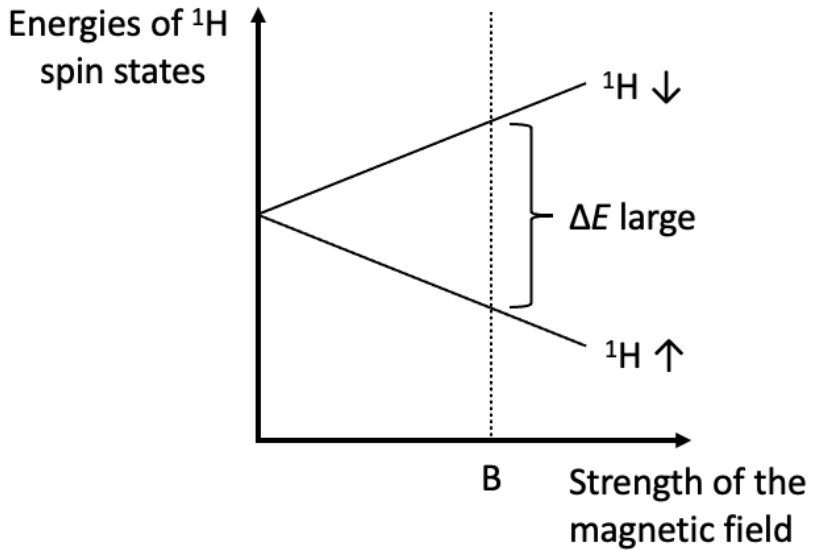


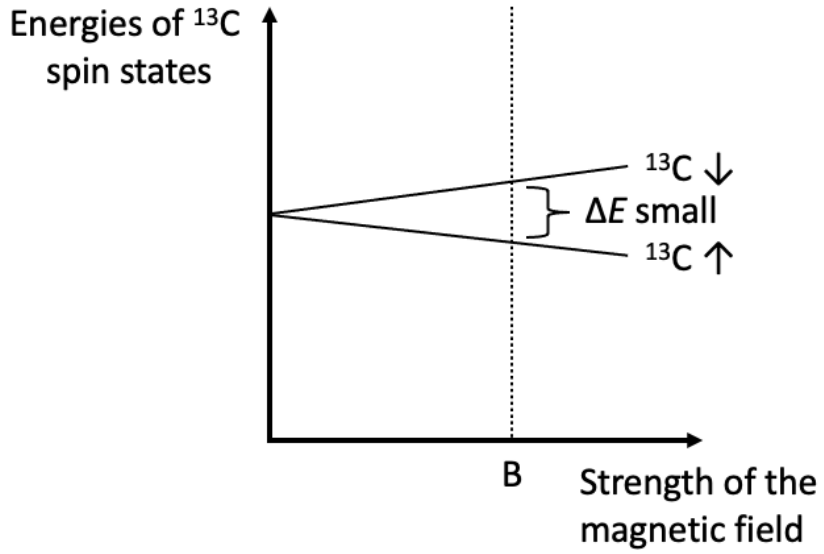
Inside the NMR Instrument



NMR – Is it even useful?!

$$\Delta E = \frac{h\gamma B_0}{2\pi}$$

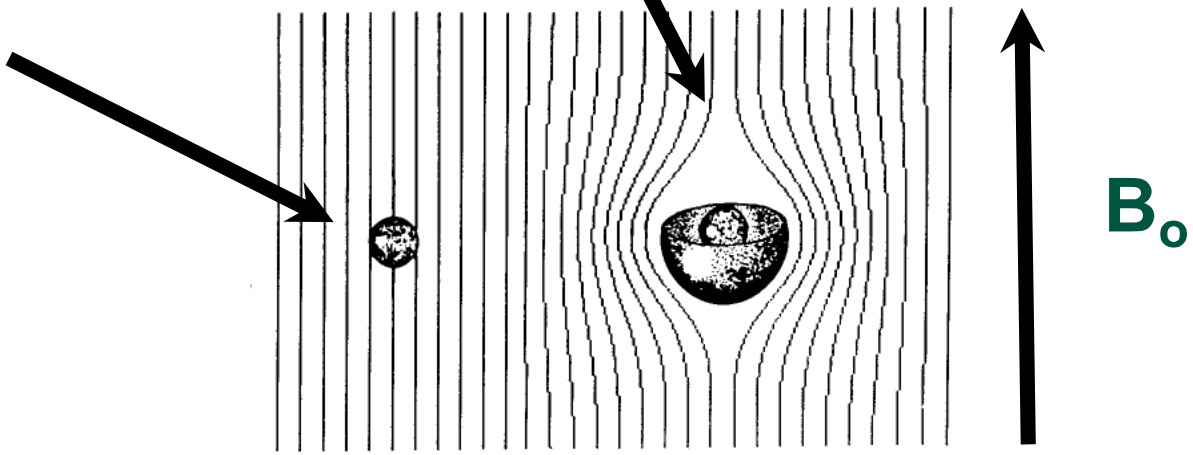




Nuclei are in different environments

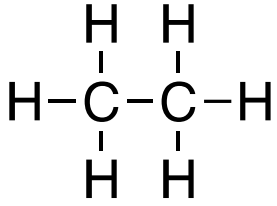
A proton (^1H) inside a molecule is shielded from B_o by the surrounding electron density : σ is (usually) **positive**

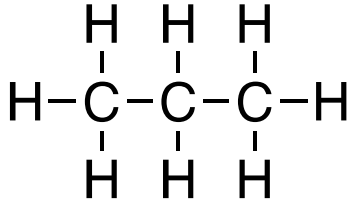
A bare proton ($^1H^+$ nucleus) has no surrounding electrons so $\sigma = 0$

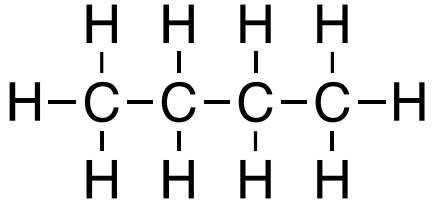


$$B_{eff} = B_o (1 - \sigma)$$

Nuclei are in different environments



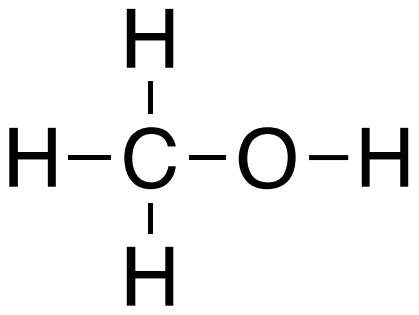


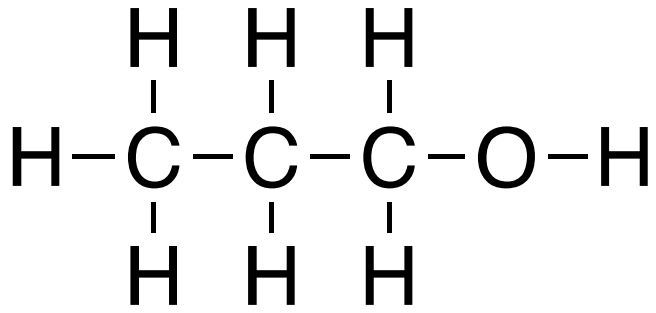


Electronegativity and Shielding

H 2.2							He
Li 1.0	Be 1.6	B 1.8	C 2.5	N 3.0	O 3.4	F 4.0	Ne
Na 0.9	Mg 1.3	Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2	Ar
K	Ca	Ga	Ge	As	Se	Br 3.0	Kr
Rb	Sr	In	Sn	Sb	Te	I 2.7	Xe

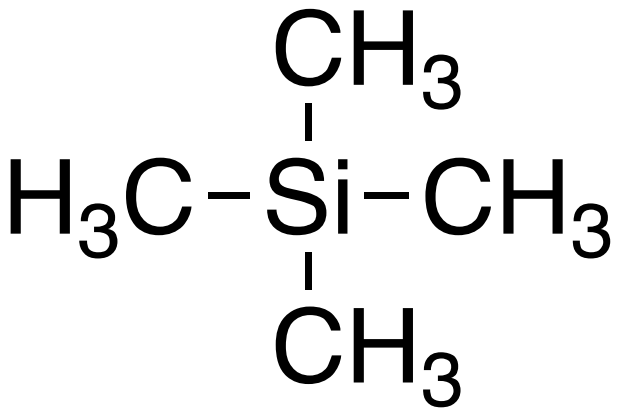
Electronegativity, Inductive Effects, and Shielding





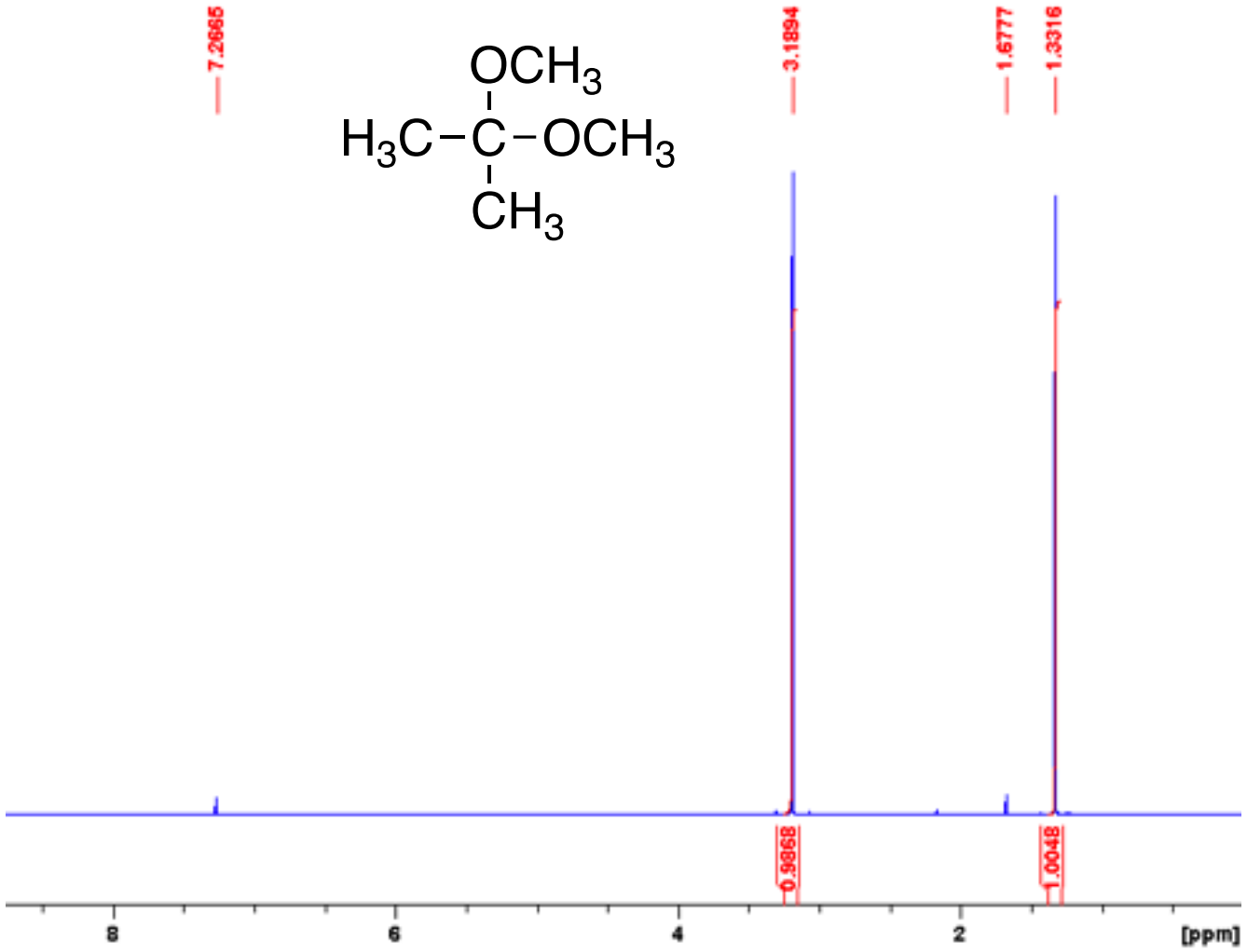
Chemical Shifts (δ)

$$\delta \quad (\text{ppm}) = \frac{\nu_{sample} - \nu_{ref}}{\nu_{ref}} \text{ x } 10^6$$



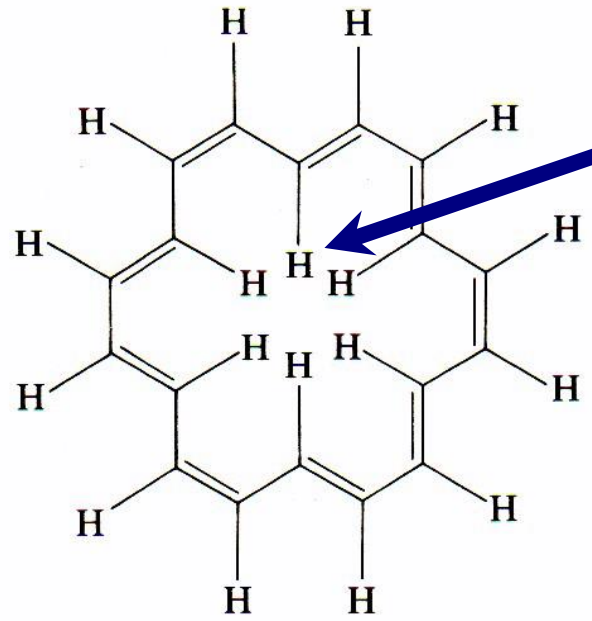
The x-axis in NMR

An NMR Example



Functional Group Magnetic Fields

Aromatic Systems can have Negative δ values

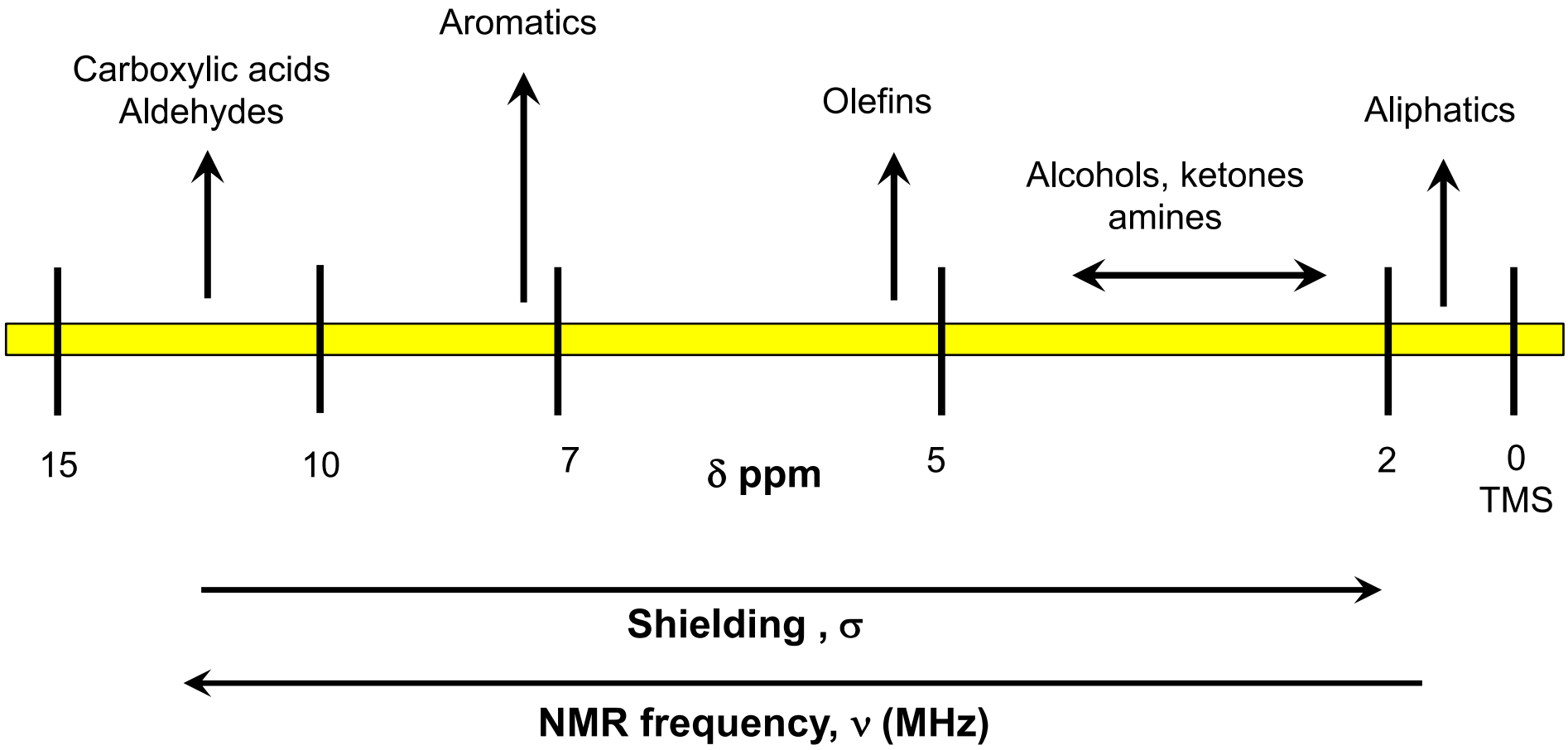


9.28 ppm

- 2.99 ppm

annulene $C_{18}H_{18}$

Typical Chemical Shifts



Chemical Shift Tables

Table 13.3 Typical chemical shift values for methyl, methylene, and methine hydrogens

Methyl hydrogens		Methylene hydrogens		Methine hydrogens	
Group*	Typical δ/ppm†	Group*	Typical δ/ppm†	Group*	Typical δ/ppm†
C–C–CH_3	0.9	C–CH_2–C	1.4	C–CH–C	1.5
R–C(=O)–C–CH_3	1.1	R–O–C–CH_2–C	1.5	R–O–CH–C	2.0
R–O–C–CH_3	1.3	C(=O)–O–C–CH_2–C	1.6	O–C(=O)–CH–C	2.5
C=C–CH_3	1.6	O–C(=O)–CH_2–C	2.2	C(=O)–CH–C	2.7
Br–C–CH_3	1.8	C=C–CH_2–C	2.3	N–CH–C	2.8
O–C(=O)–CH_3	2.0	C(=O)–CH_2–C	2.4	Ar–CH–C	3.0
C–C(=O)–CH_3	2.2	N–CH_2–C	2.5	Ar–C(=O)–CH–C	3.4
Ar–CH_3	2.3	Ar–CH_2–C	2.7	R–O–CH–C	3.7
N–CH_3	2.3	Ar–C(=O)–CH_2–C	2.8	HO–CH–C	3.7
Ar–C(=O)–CH_3	2.5	R–O–CH_2–C	3.4	Br–CH–C	4.3
C(=O)N–CH_3	2.9	Br–CH_2–C	3.4	ArO–CH–C	4.5
R–O–CH_3	3.3	Cl–CH_2–C	3.5	C(=O)O–CH–C	4.8
C(=O)O–CH_3	3.7	HO–CH_2–C	3.6	Ar–C(=O)O–CH–C	5.1
Ar–O–CH_3	3.8	Ar–O–CH_2–C	4.0		
		C(=O)–O–CH_2–C	4.1		

* R indicates an alkyl group; Ar indicates an aromatic group.

† Typically, resonance signals are within ±0.2 ppm of the values quoted.

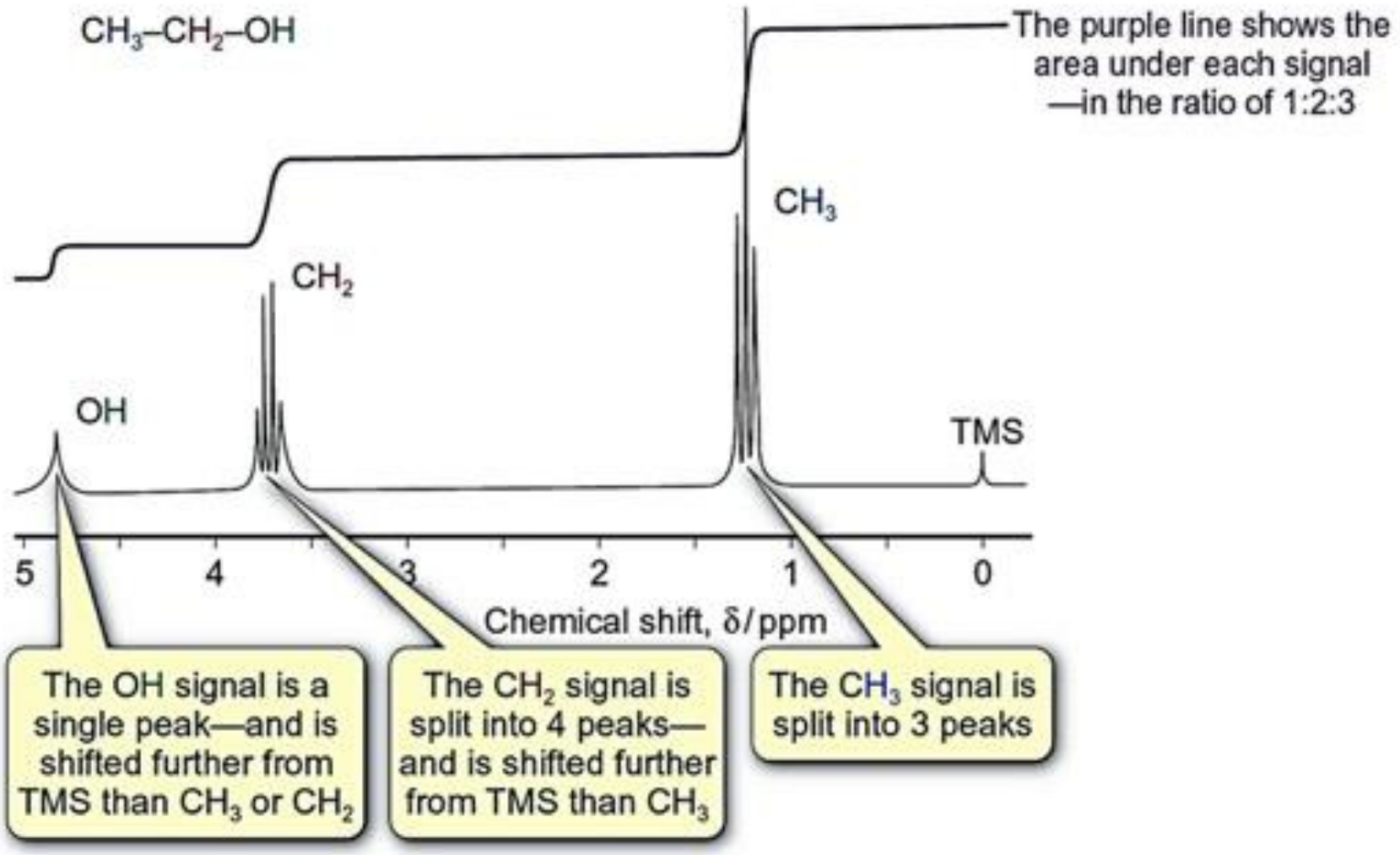
Chemical Shift Tables

Type of proton	Approximate δ/ppm	Type of proton	Approximate δ/ppm
$(CH_3)_4Si$	0	Aromatic *H*	6.5 – 8
CH_3–	0.9	RC(O)–*H* (aldehyde)	9.0 – 10
–CH_2–	1.3	I–C–*H*	2.5 - 4
–C*H*–	1.4	Br–C–*H*	2.5 - 4
C=C-CH_3	1.7	Cl–C–*H*	3 - 4
C_6H_5–CH_3	2.1	F–C–*H*	4 - 4.5
-C≡C–*H*	2.3	RNH_2	Variable 1.5-4
R–O–CH_3	3.3	RO*H*	Variable 2-5
R_2C=CH_2	4.7	ArO*H*	Variable 4-7
R_2C=C*H*R	5.3	RC(O)O*H*	Variable 10-12
CH_3-C(O)–	2-2.5		

Key Features of NMR Spectra

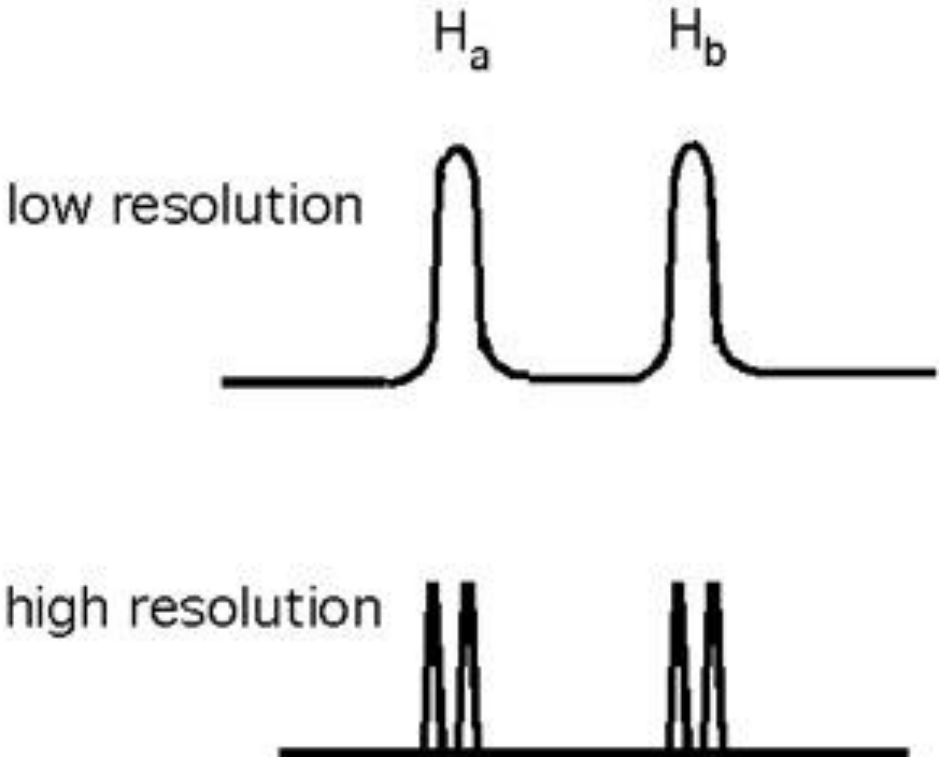
- Number of peaks
- Chemical Shift
- Integrals
- *Coupling*

Magnetic Coupling Effects and Multiplet Splitting

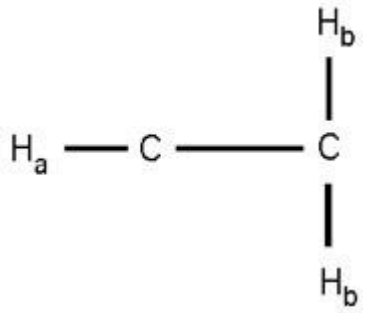


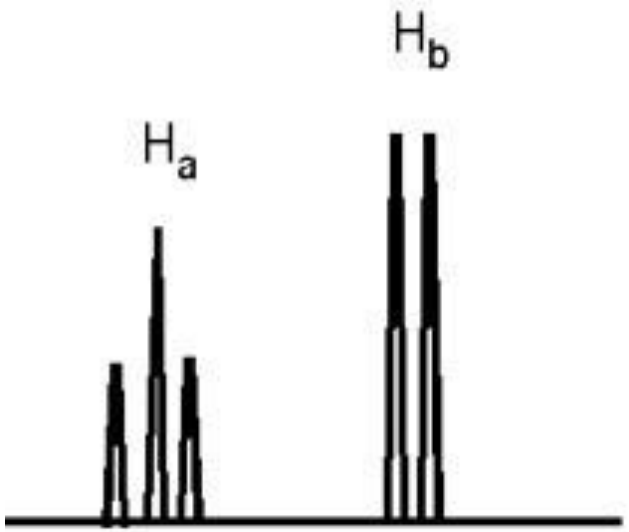
Spin-Spin Coupling : How Multiplet Patterns Arise

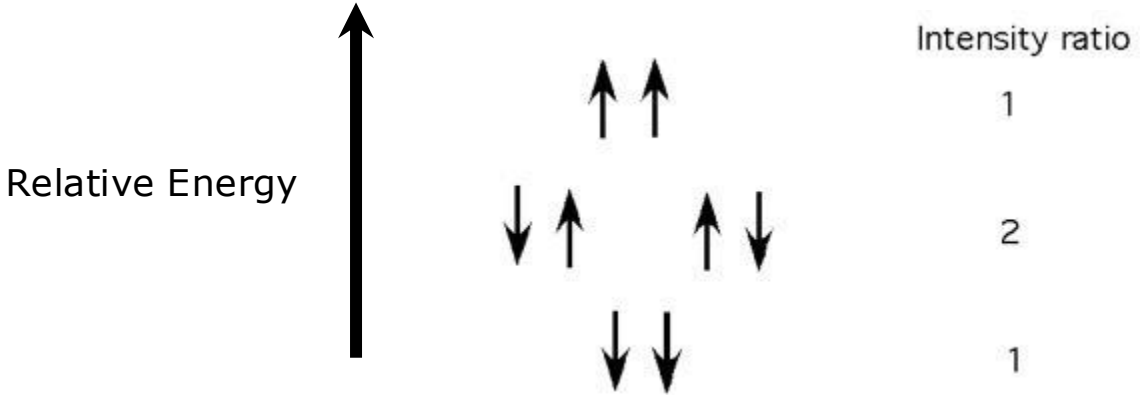




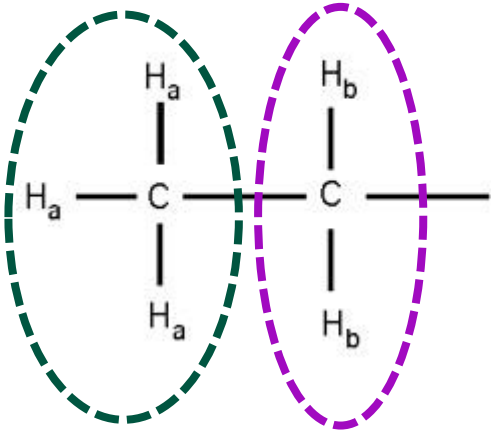
Forming a Triplet Pattern

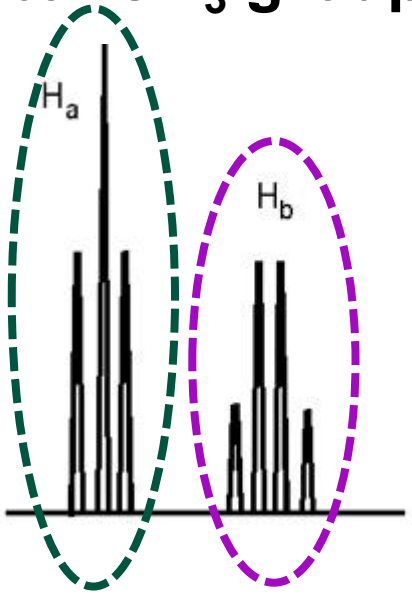


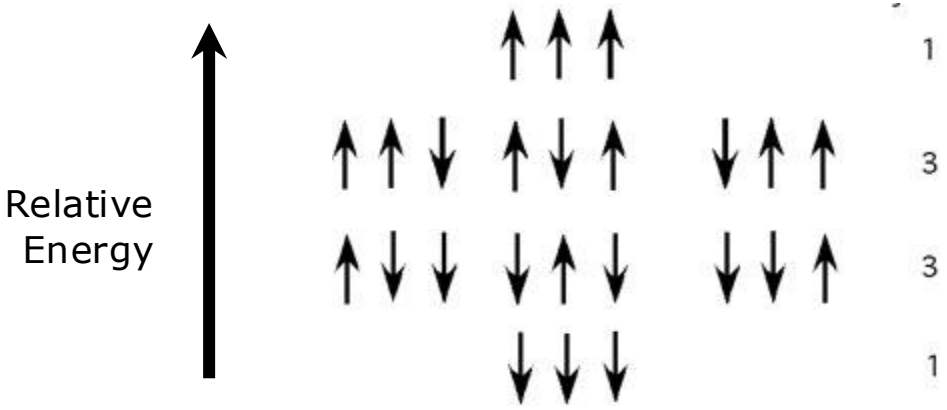




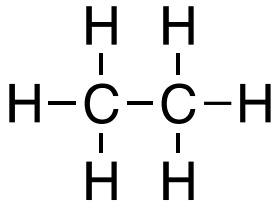
Forming a Quartet : coupling to -CH_3 groups

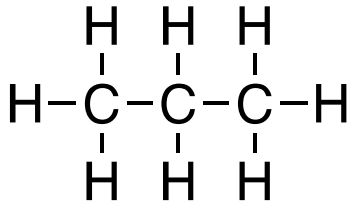


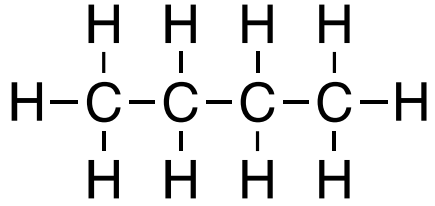




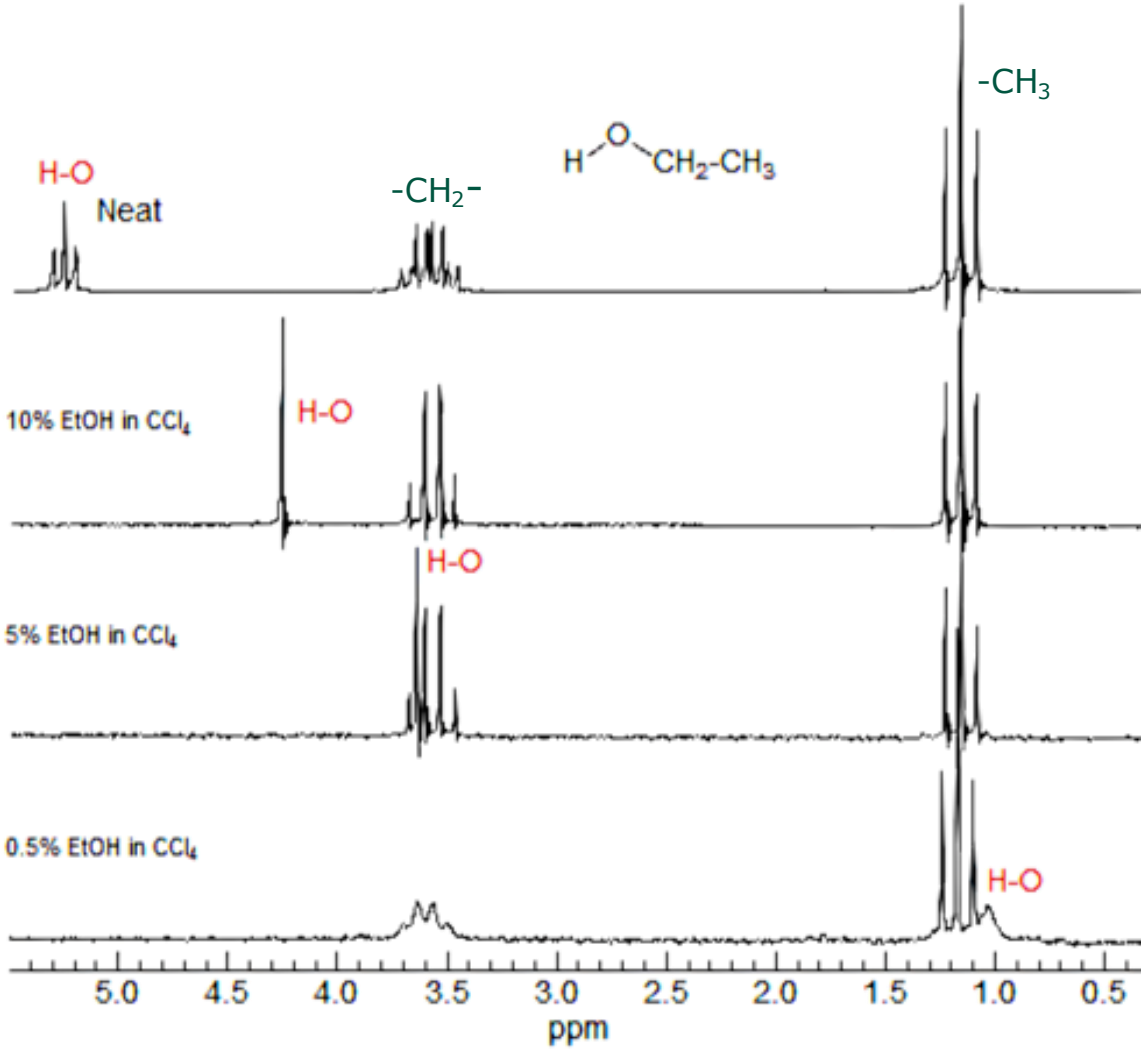
Equivalent protons do not couple each other



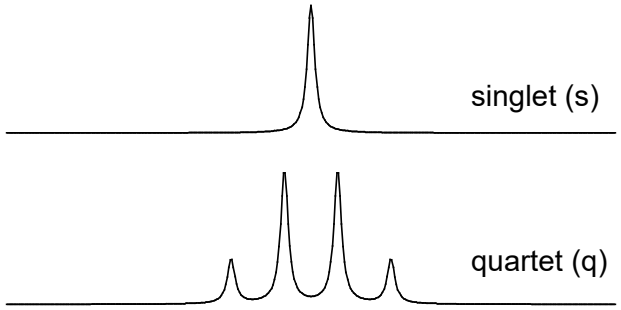


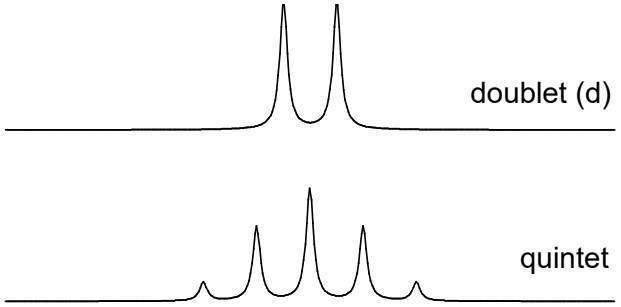


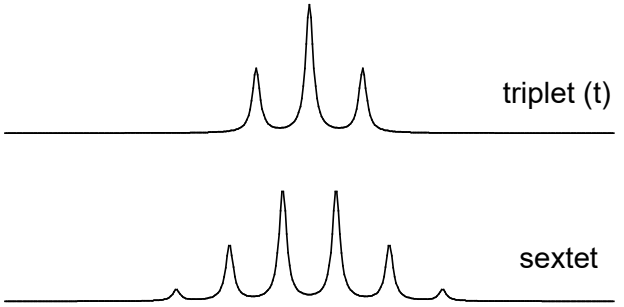
What about O-H, N-H protons ?

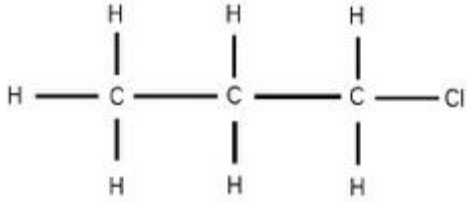


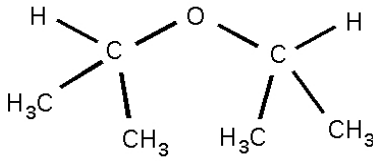
Simple multiplet patterns

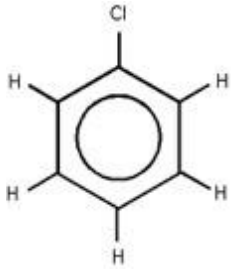


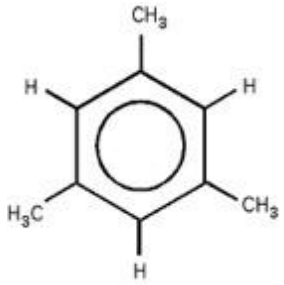






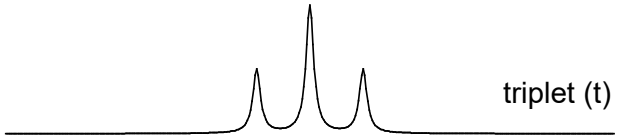


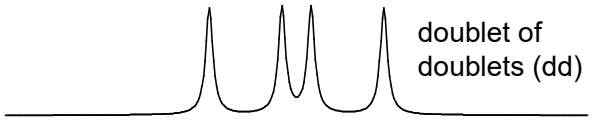




More examples and test-type questions are provided in Chemistry[3]

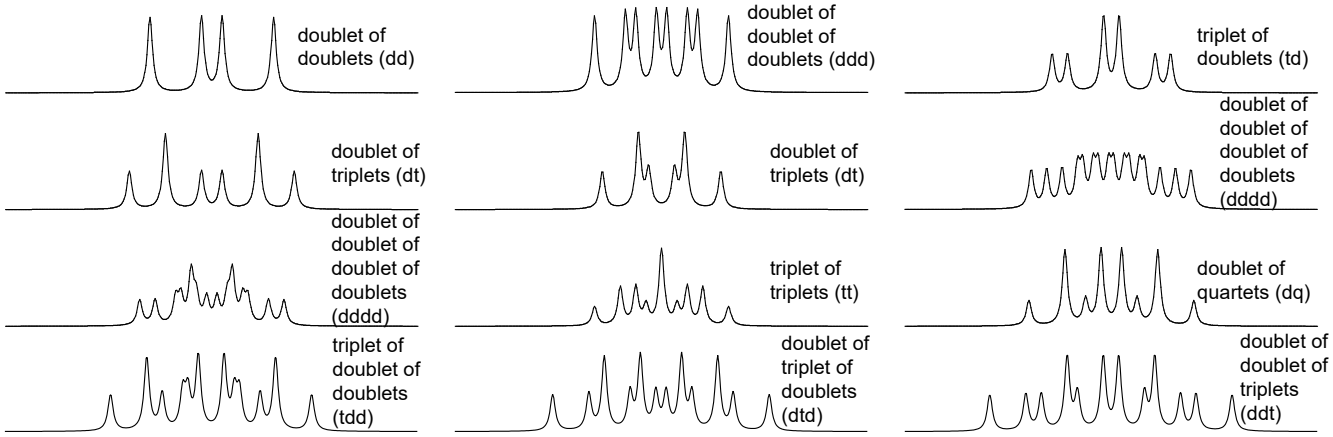
Simple vs. Complex





More complex multiplet patterns

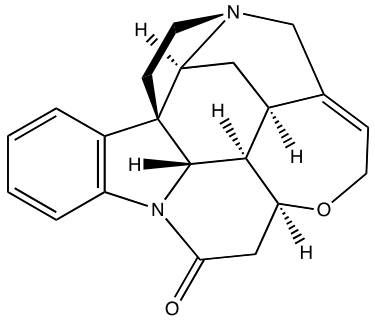
We can also see more complex "higher order" multiplet patterns caused by magnetic coupling to two (or more) different sets of protons in different environments. This gives rise to patterns such as: "doublets of doublets", "doublets of triplets", "triplet of doublets" etc.



For example, try to predict the multiplet pattern for 3,3 dimethyl-1-butene : CH_3-CH=CH_2

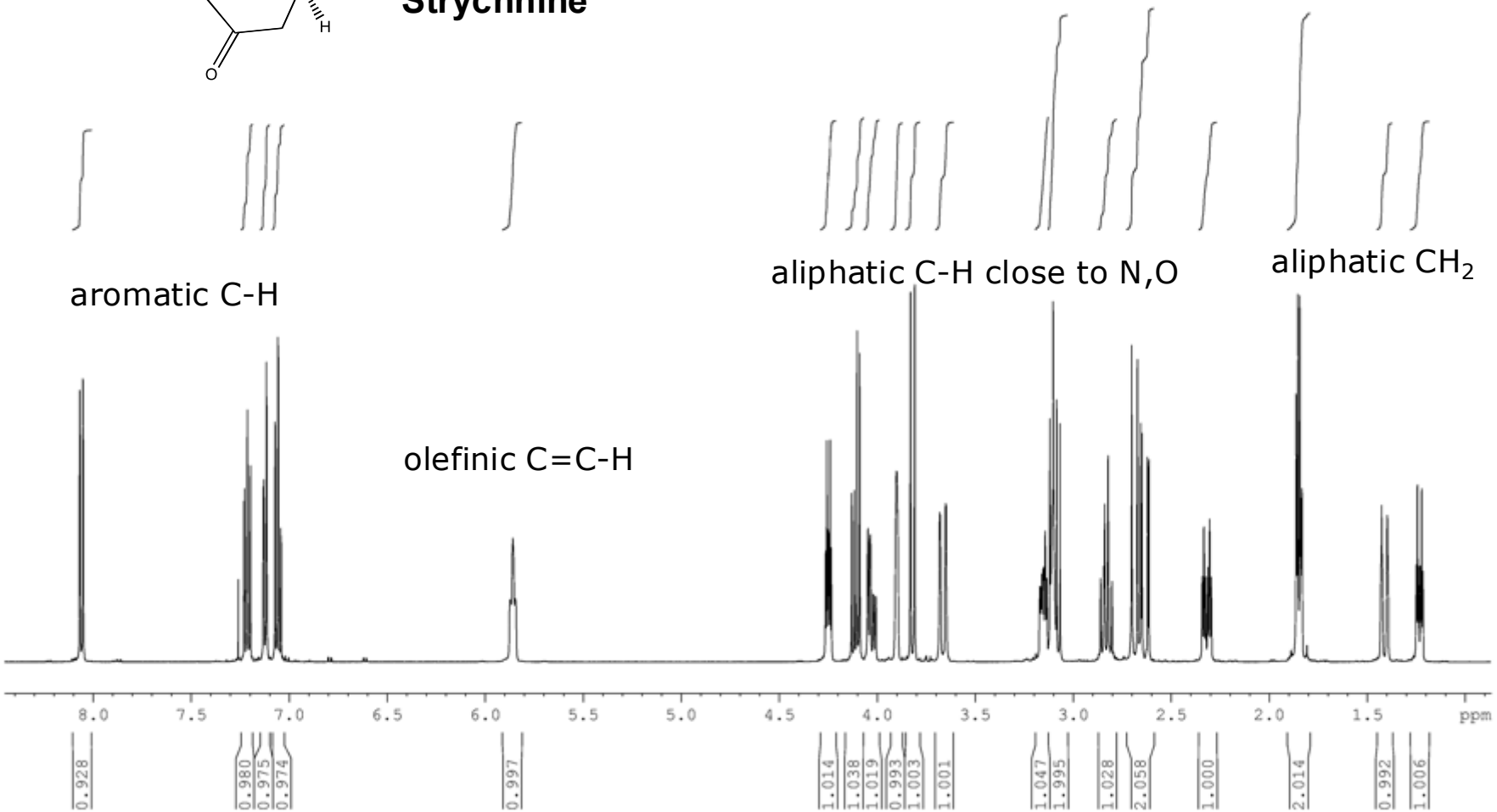
Useful websites : http://www.nmrdb.org/predictor/

Online database : http://sdbs.db.aist.go.jp/sdbs/cgi-bin/cre_index.cgi



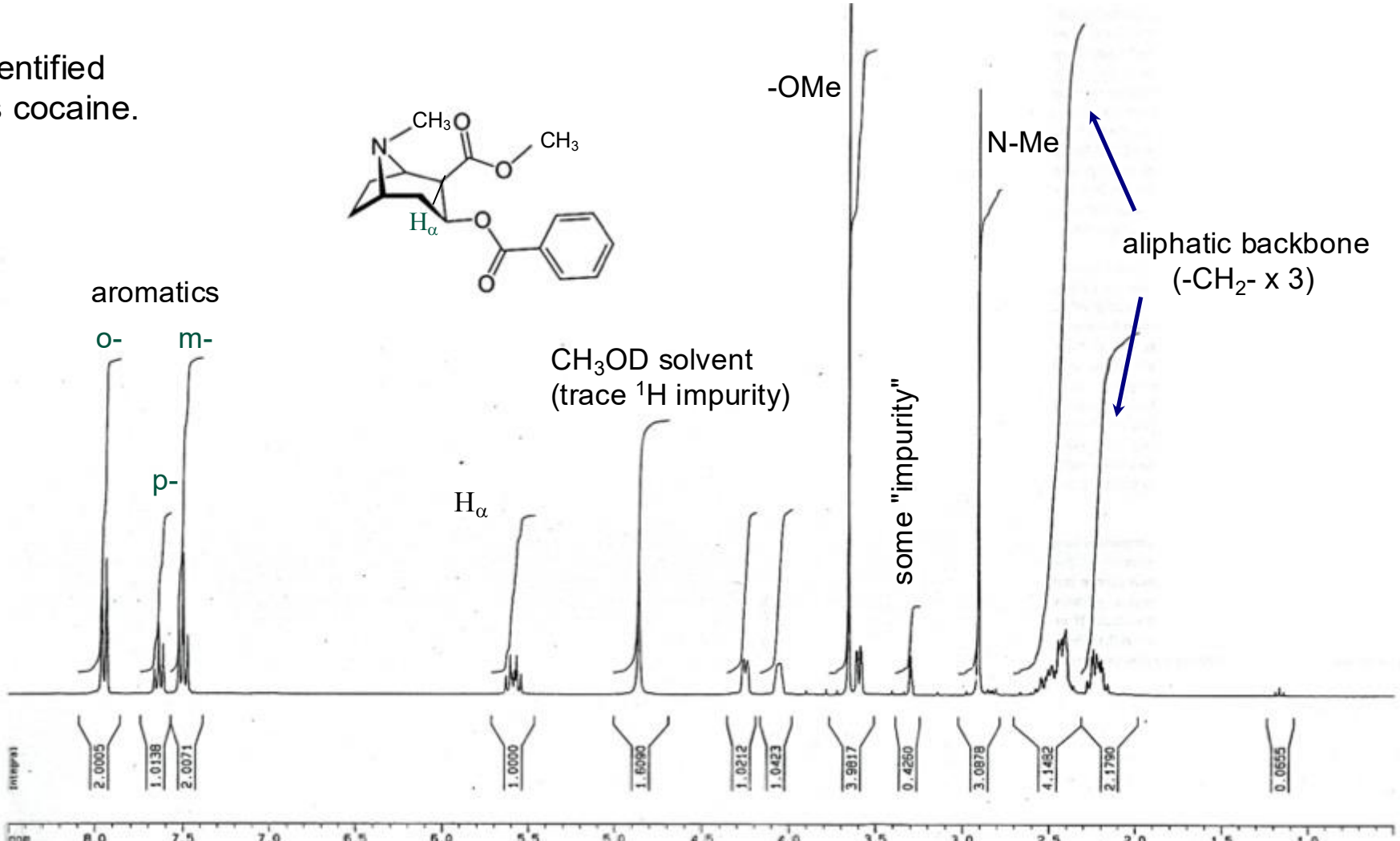
Strychnine

Interpreting NMR spectra of larger molecules can quickly become complicated: can you make sense of the peak assignments here ?



However, we can also use NMR spectra as a "fingerprint" to identify an unkown substance, by comparing with known compounds. For example, an unlabelled white powder found in a cupboard of a Chemistry department ...

Identified as cocaine.



Using spectroscopy to determine molecular structure

- Chemical composition (Elemental analysis, Mass spectrometry, Infrared spectroscopy)

- Look at the ^{1}H NMR spectrum.
  - Kinds of environment (number of signals)
  - Integrals (number of nuclei in each environment)
  - Shift (chemical shift tells about environment)
  - Splitting (multiplet – information about neighbors)

- Start to build possible models of the whole structure, or fragments contained within it.

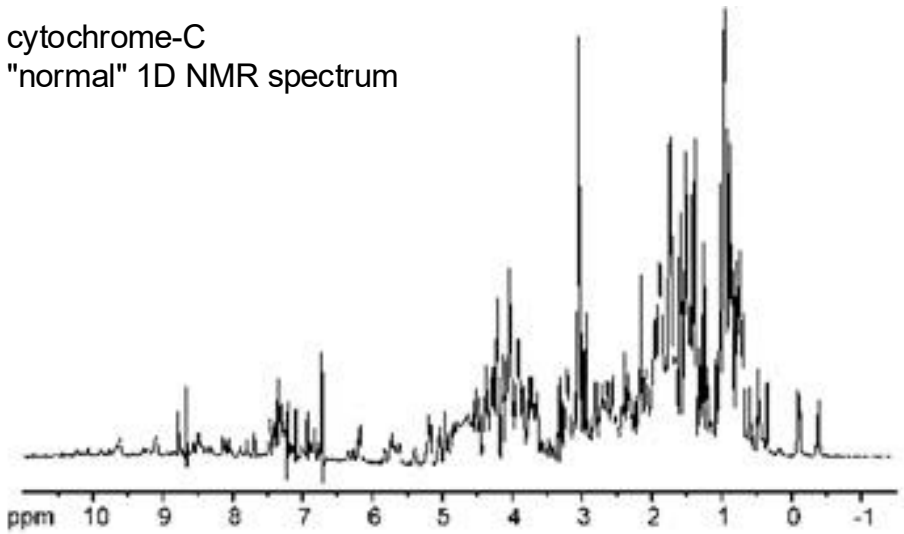
- Check your structure against the data

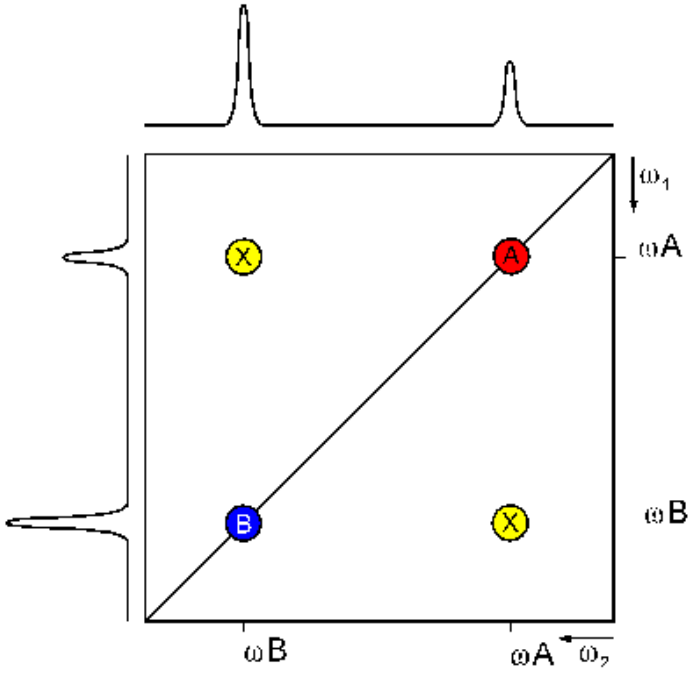
- If you have more than one model that fits all the data, suggest additional experiments.

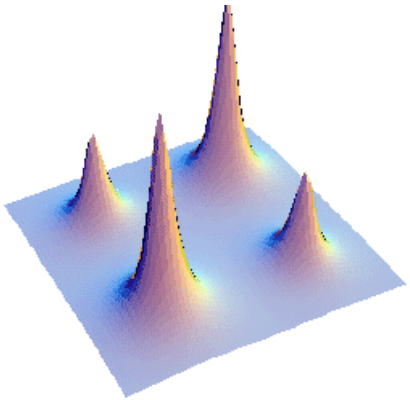
see Chemistry^{3} for some worked examples and possible test questions

Complex Biomolecular Structures : "Two-Dimensional" (2-D) NMR

cytochrome-C
"normal" 1D NMR spectrum







Some multidimensional NMR pulse sequences and their "interesting" acronyms

COrrelation **S**pectroscop**Y (COSY)**

TOtal **C**orrelation **S**pectroscop**Y (TOCSY)**

Nuclear **O**verhauser **E**ffect **S**pectroscopY (**NOESY**)

Heteronuclear **S**ingle **Q**uantum **C**orrelation (**HSQC**) Spectroscopy

Heteronuclear **M**ultiple **B**ond **C**orrelation (**HMBC**) Spectroscopy

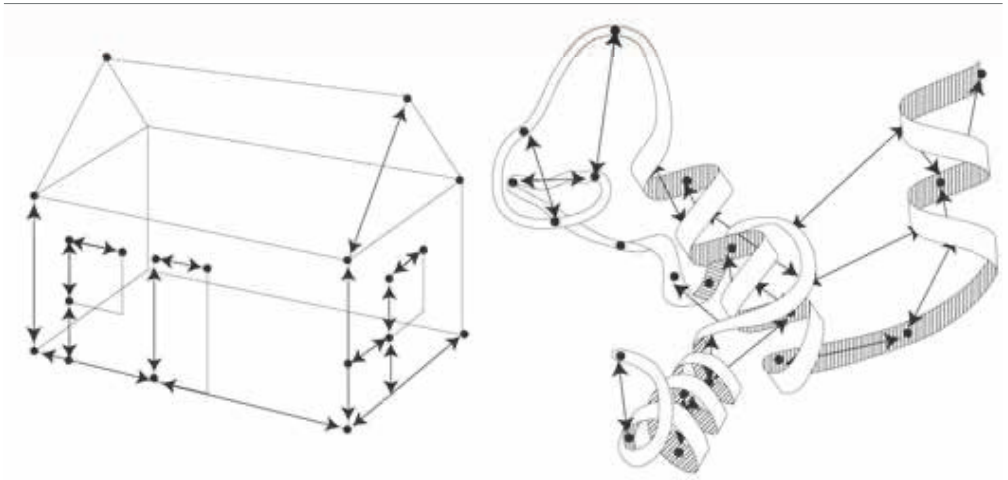
Incredible **N**atural **A**bundance **D**oubl**E Qua**ntum **T**ransfer **E**xperiment **(INADEQUATE)**

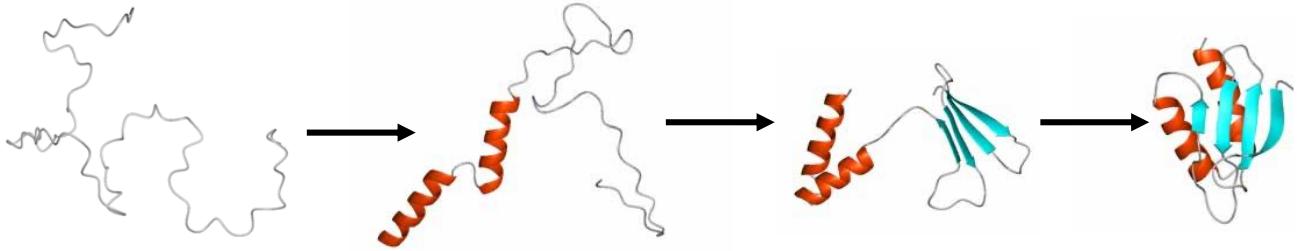
Polarization **E**nhancement **N**urtured **D**uring **A**ttached **N**ucleus **T**esting **(PENDANT)**

Delays **A**lternating with **N**utation for **T**ailored **E**xcitation **(DANTE)**

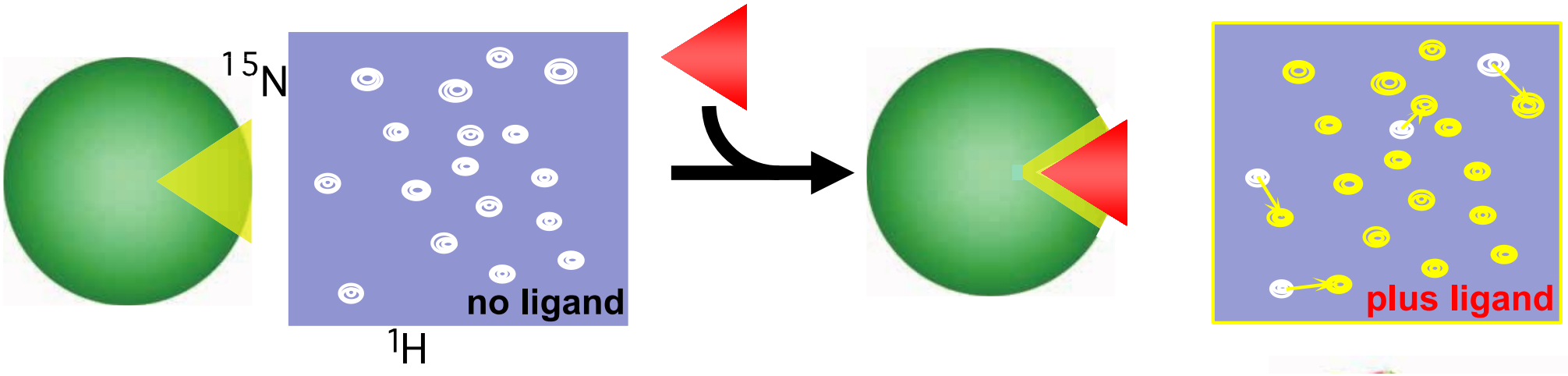
Protein Structure Determination by NMR

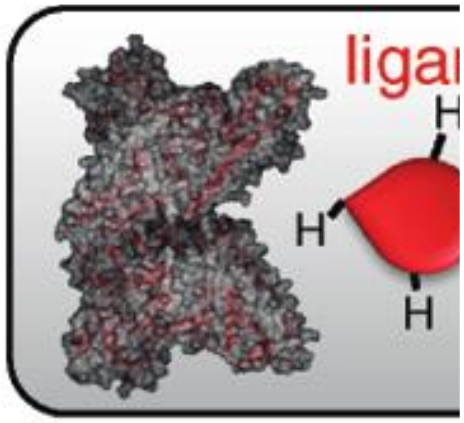
- Often requires ‘labelling’
- Protein structure is studied in solution
- Allows investigation of dynamics





The 'Environment' can be more than one molecule

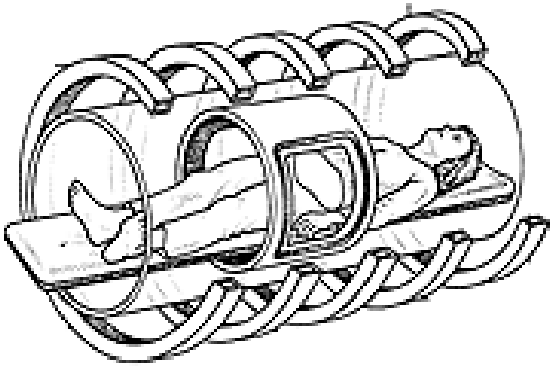


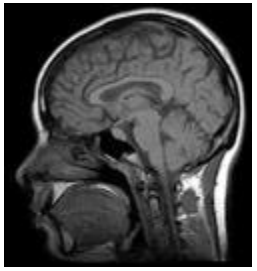


MRI (Magnetic resonance imaging)

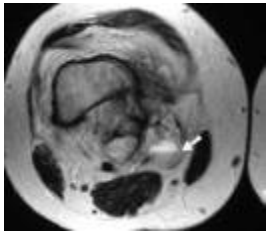


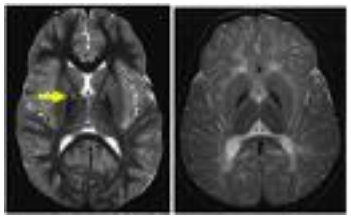
MRI is based on looking at the protons (^{1}H nuclei) of H_2O molecules – i.e. water distributed throughout soft tissue rather than dense skeletal bone. Use **pulsed NMR** to observe the "relaxation time" of protons oriented by the magnetic field –we distinguish "longitudinal" (T_1) and "transverse" (T_2) relaxation times. Different regions are distinguished by the dynamic relaxation behaviour of the water molecules. We use **tomography** techniques to study slices at different depths within the tissue - 3D images. Gd^{3+} complexes are often injected as contrast agents to speed up T_1 and "light up" specific regions.







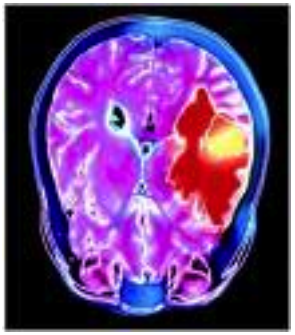




"Functional" MRI (fMRI) and PET Imaging Techniques



Unlike MRI that is based on ^{1}H relaxation, fMRI looks at the magnetic spin state of Fe (iron) in oxygenated hemoglobin during increased O_2 consumption – it is used to map brain activity in real time



PET scans : positron emission tomography : radioactive ^{15}O : decays (122 s) by e^{+} emission.

PET scan images are used to map brain activity on second timescales in patients injected with ^{15}O tracer.

